

JobRadar Case Study

Job Radar

Project Overview

Job Radar is a Python-based job search automation tool designed to reduce the time and effort required to review large volumes of job alert emails. The application connects directly to Gmail, extracts job opportunities from job-alert providers, scores opportunities based on user-defined criteria, and surfaces the most relevant opportunities for review.

The project was created to solve a personal challenge: manually reviewing dozens of job-alert emails every day from multiple sources while searching for Salesforce Administration, Business Analysis, Sales Operations, RevOps, and Data Analytics opportunities.

Business Problem

Job seekers often receive a high volume of job alert emails from platforms such as LinkedIn, Indeed, CareerBuilder, and Google Alerts.

A single LinkedIn alert may contain multiple job opportunities, resulting in hundreds of potential openings being delivered each week.

Challenges included:

- Job alerts scattered across multiple email providers
- Duplicate opportunities across platforms
- Manual review of dozens of emails per day
- Difficulty prioritizing the most relevant opportunities
- Time spent reviewing poor-fit roles

The existing process required manually opening each email, reviewing each position, and deciding whether it aligned with career goals.

Solution

Job Radar connects directly to Gmail using the Google Gmail API and OAuth authentication.

The application:

1. Retrieves job-alert emails from approved job sources
2. Extracts job opportunities from email content
3. Parses job titles, companies, and locations
4. Scores opportunities based on predefined criteria
5. Exports ranked results into a structured CSV file for review

The goal is to transform a cluttered inbox into a prioritized list of opportunities.

MVP Features Delivered

Gmail Integration

- OAuth authentication
- Secure Gmail API connection
- Read-only email access

Job Alert Filtering

Filters job-related emails from:

- LinkedIn Job Alerts
- CareerBuilder
- Indeed
- Google Alerts

Job Extraction

Extracts:

- Job Title
- Company
- Location
- Source
- Email Date

Match Scoring Engine

Initial scoring based on:

- Salesforce keywords
- CRM keywords
- Business Analyst keywords
- RevOps keywords
- Location matching
- Remote and Hybrid indicators

CSV Export

Exports opportunities to:

reports/job_radar_results.csv

allowing further review and analysis in Excel.

Tools & Technologies

- Python
 - Gmail API
 - OAuth 2.0
 - Google Cloud Platform
 - Pandas
 - BeautifulSoup
 - CSV Export
 - VS Code
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Skills Demonstrated

Business Analysis

Identified a real-world workflow problem and translated it into a technical solution.

Process Automation

Automated manual job-search activities.

API Integration

Connected an application to Gmail using OAuth authentication and the Gmail API.

Data Processing

Extracted, transformed, and organized unstructured email data.

Data Storytelling

Converted large volumes of email data into actionable job-search insights.

Product Thinking

Designed a solution focused on reducing decision fatigue and improving efficiency for job seekers.

Results

The MVP successfully:

- Connected to Gmail
- Retrieved job-alert emails
- Extracted job opportunities
- Generated opportunity scores
- Exported structured results for analysis

The project transformed job-search emails into a centralized opportunity review process.

Future Enhancements

V1.2

- Improved job extraction logic
- Removal of non-job content
- Enhanced scoring engine
- Remote and hybrid prioritization

V2

- AI-powered job fit analysis
- Resume matching
- Salary estimation
- Duplicate detection

V3

- Automated application tracking
 - Recruiter outreach management
 - Interview pipeline management
 - Personalized opportunity recommendations
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Key Lesson Learned

The most valuable insight from this project was that the problem was not finding jobs—it was filtering and prioritizing opportunities efficiently.

By automating the intake and scoring process, Job Radar shifts the focus from searching for jobs to evaluating the best opportunities.

JobRadar V1.2

Built a Python application that connects to Gmail using the Gmail API and automatically processes job alert emails from LinkedIn, Google Alerts, CareerBuilder, and other job sources.

The application extracts job opportunities, removes duplicates, scores each role against predefined career goals, categorizes opportunities by career track, and generates recommendation reports.

Features include automated Gmail ingestion, job extraction, duplicate removal, career track classification, scoring logic, and report generation.

Outputs include Apply Now, High Priority, Review, and Skip reports to reduce manual job search effort and focus attention on the most relevant opportunities.

Technologies used include Python, Gmail API, OAuth 2.0, Pandas, CSV reporting, and rule-based scoring logic.

V1.3 Enhancements

- Added career track classification
- Added recommended resume selection
- Added match percentage scoring
- Added application prioritization logic
- Improved decision support for job seekers